



Customer Retention & Churn Analysis



Objective

The objective of this project is to analyze customer churn patterns, identify factors affecting customer retention, and provide recommendations to reduce customer loss and improve customer satisfaction.



Dataset Description

The dataset contains customer information including:

- Customer ID
- Gender
- Contract Type
- Internet Service
- Payment Method
- Monthly Charges
- Total Charges
- Churn Status



Tools Used

- Python
- Pandas
- Matplotlib
- VS Code



Analysis Performed

1. 👥 Customer Churn Overview

💰 Total Customers: 7043

✅ Retained Customers: 5174

❌ Churned Customers: 1869

📊 Churn Rate: 26.54%

2. 👤 Gender-wise Churn Analysis

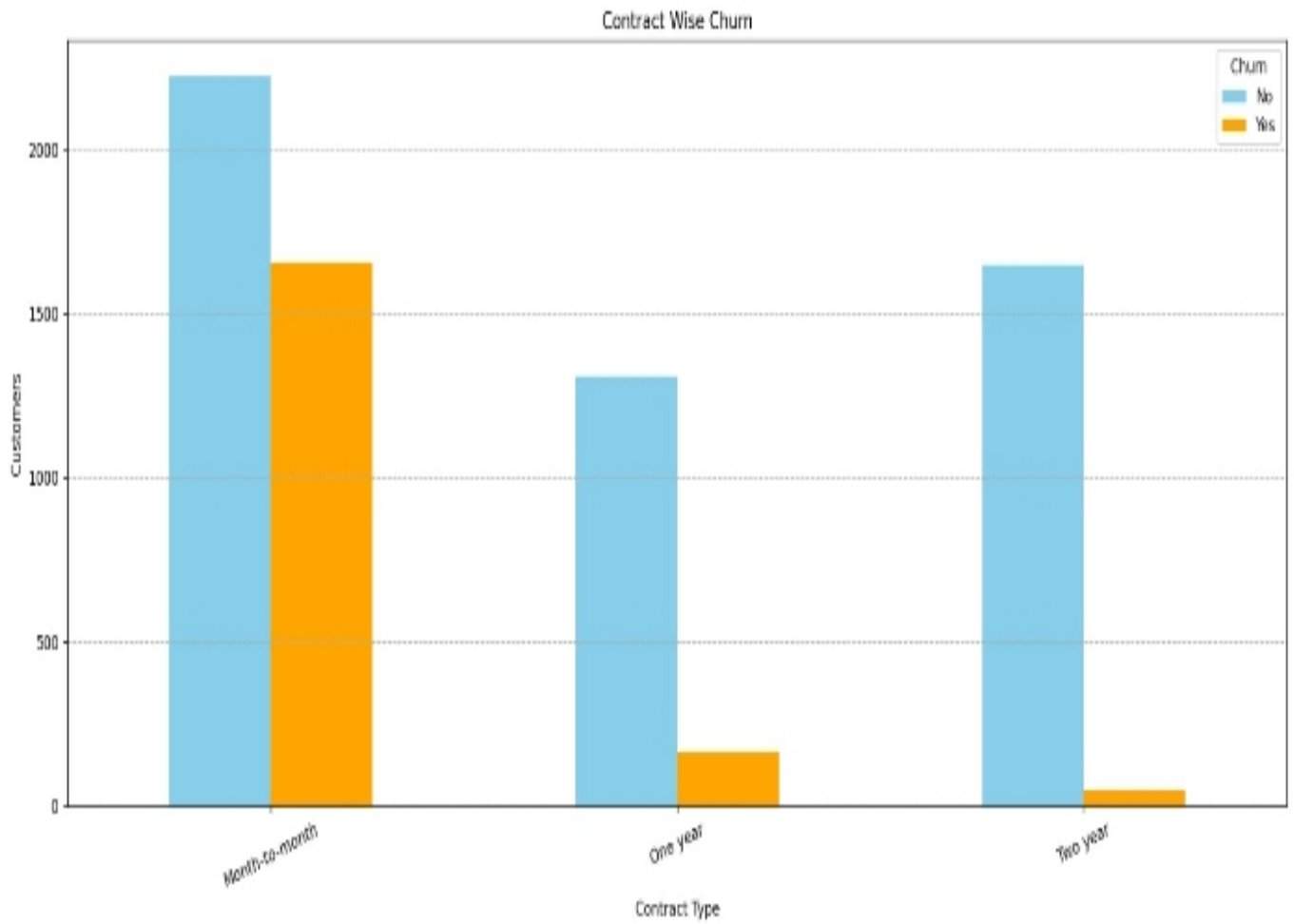
Gender	Churned Customers
Female	939
Male	930

📌 **Observation:** Customer churn is almost equally distributed between male and female customers.

3. 📄 Contract-wise Churn Analysis

Contract Type	Churned Customers
Month-to-Month	1655
One Year	166
Two Year	48

📌 **Observation:** Customers with month-to-month contracts are significantly more likely to churn.



4. 🌐 Internet Service-wise Churn Analysis

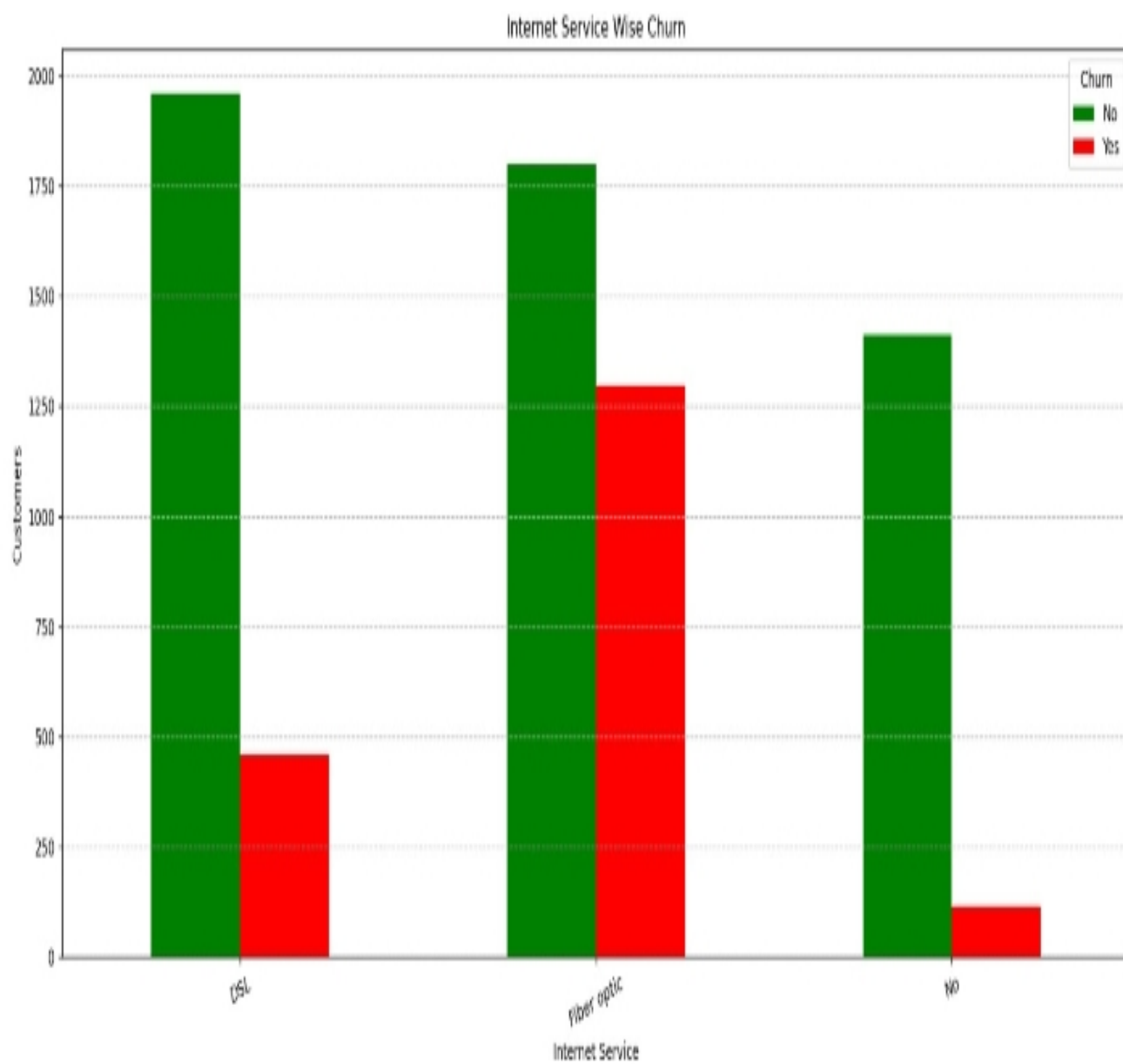
Internet Service Churned Customers

Fiber Optic 1297

DSL 459

No Internet Service 113

📌 **Observation:** Fiber Optic customers recorded the highest churn rate.



5. 🏠 Payment Method-wise Churn Analysis

Payment Method Churned Customers

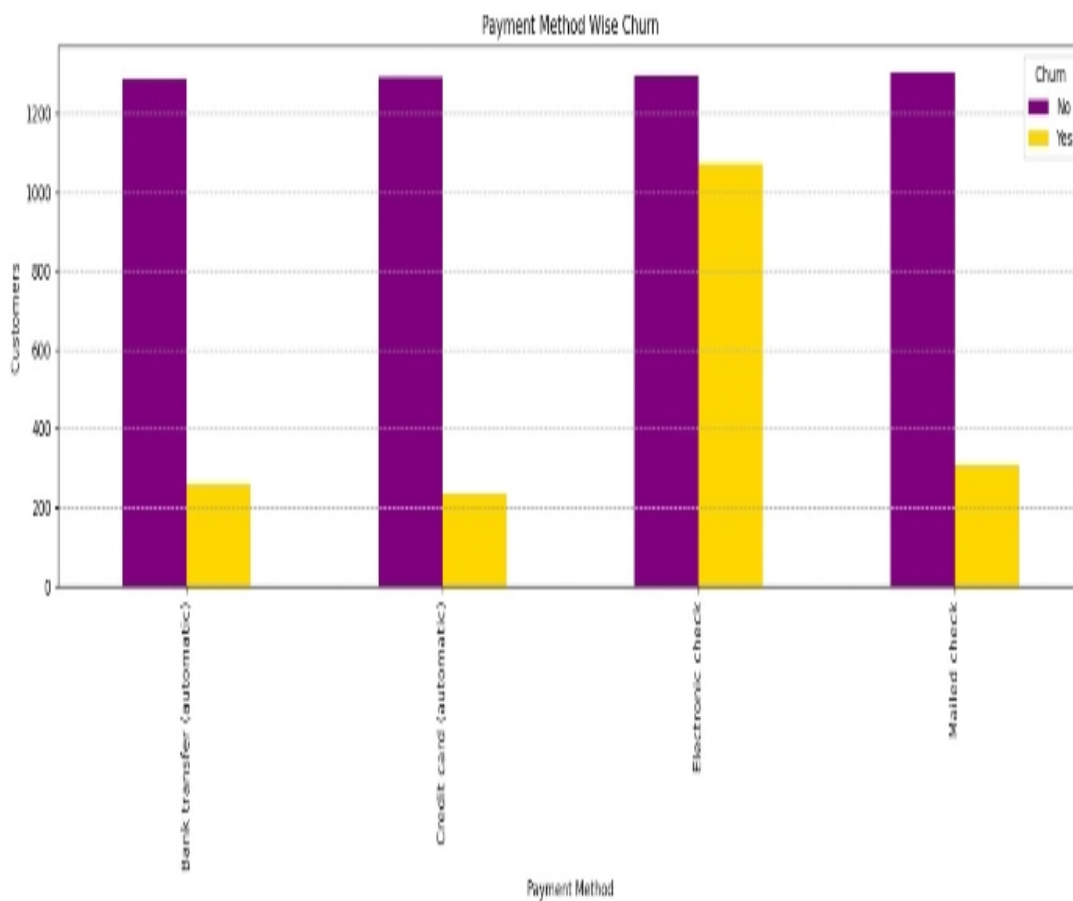
Electronic Check 1071

Mailed Check 308

Bank Transfer (Automatic) 258

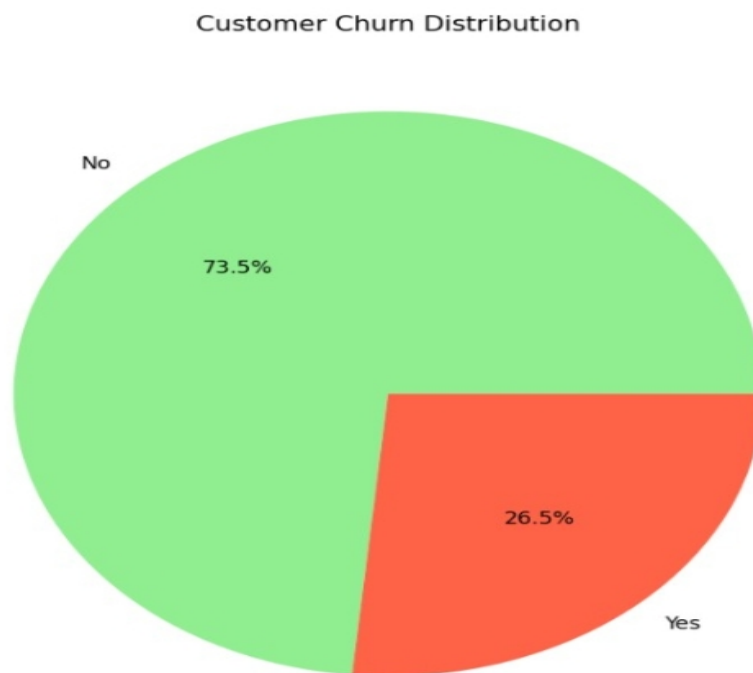
Credit Card (Automatic) 232

📌 **Observation:** Customers using Electronic Check showed the highest churn rate.



Customer Churn Distribution

 The overall churn rate is 26.54%, while 73.46% of customers were retained.



Key Findings

 Total customer base consists of 7043 customers

- ✅ Overall churn rate is 26.54%
- ✅ Month-to-month contracts have the highest churn
- ✅ Fiber Optic customers are more likely to leave the service
- ✅ Electronic Check users show the highest churn behavior

💡 Recommendations

Encourage customers to switch to long-term contracts.

Improve service quality for Fiber Optic customers.

Promote automatic payment methods.

Provide loyalty rewards and retention programs.

Monitor high-risk customer segments proactively.

🏁 Conclusion

This analysis identified major factors influencing customer churn. By improving customer experience, encouraging long-term contracts, and implementing effective retention strategies, businesses can reduce churn and improve customer loyalty.

👤 Author

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